

A Battery Pack Reconfiguration Scheme for Improved Charge Balancing

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Abstract—Charge balancing is an important consideration for Lithium-ion battery management systems in protecting cell health and maximizing performance. Many existing balancing systems employ passive balancing during charging cycles by resistive dissipation, or use active balancing with switched-capacitor schemes or DC-DC converters to drive balancing between cells. Methods for active balancing are generally more efficient than those for passive balancing and can be used more frequently, but are more complex and costlier. Through laboratory experimentation, this article investigates an innovative idea for pack reconfiguration to improve cell balancing that has the simplicity of passive balancing with the efficiency characteristic of active balancing schemes. The idea is centered around a grid configuration pack, reconfigured through a simple switching network to a parallel configuration. During periods of charging in the parallel configuration, post-charge results show improvement in terms of pack balance as compared to charging in the full grid configuration. Self-balancing with no measurable energy loss was shown to achieve a small imbalance over the course of a resting period when starting with an imbalanced pack.

I. INTRODUCTION

Lithium-Ion battery technology is becoming the industry standard for electric vehicle (EV) energy sources due to the high energy density and relatively high power density of the various chemistries. The use of Li-Ion limits cell voltages to under 5V, necessitating the combination of many Li-Ion cells into packs to provide the higher voltages required for automotive drivetrains [1]. Adding more cells in parallel increases the pack capacity, thus typical configurations combine modular structures of cells in series and parallel. The nomenclature used to characterize the structure of such a pack has the format $xPyS$ – where x is the number of parallel modules composed of y cells in series – or $xSyP$ – where x is the number of series-connected modules composed of y cells in parallel, known also as “grid configuration”, the focus of this article.

Parameters and characteristics of each individual cell in a pack made up of “identical” cells will certainly vary. For example, internal (ohmic) resistance will vary or individual capacity will vary due to imperfections in the manufacturing process. After cyclic use, these variations lead to differences in charging and discharging rates among the cells. Over time, significant differences in state of charge (SoC) can develop. This limits how much a pack can be used before certain cells experience damage due to charging or discharging beyond their individual capacities.

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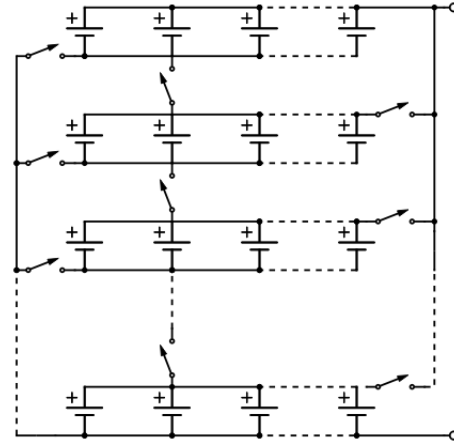


Fig. 1. Grid-to-Parallel Reconfiguration Switching Scheme

Many methods have been developed to directly address SoC imbalance with additional circuitry. Common implementations employ passive balancing, where excess energy in selective cells is dissipated through shunt resistors [2][3][4]. Such implementation is cheap and relatively simple, but inefficient from an energy perspective. That is, power losses due to passive balancing have been estimated to be as high as 544W in high-capacity automotive grade packs [5]. Other charge balancing methods actively redistribute charge throughout the battery pack using switched capacitor schemes [6], or bi-directional DC-DC converters [7][8]. These methods are more efficient in saving energy, but are more costly and require more complex control schemes.

Other methods have looked to indirectly address SoC imbalance through pack reconfiguration, bypassing specific at-risk cells and modules during operation to prevent damage [9][10].

This article investigates the use of an innovative yet simple pack reconfiguration method shown in Figure 1 for charge balancing purposes. Viewing a grid configuration pack as several parallel modules connected in series, the inner switches of Figure 1 control the serial connections of the modules. The outer switches connect the positive and negative terminals of each module to their respective bus, effectively creating a single parallel configuration of cells. The total number of switches required is $3(n-1)$, where n is the number of modules connected in series in the pack. This number of switches compares favorably to other suggested active and passive balancing schemes.

When in parallel configuration, the higher voltage cells will naturally discharge energy into the lower voltage cells

until there is a uniform voltage. Because of complexity and uncertainty in testing environments, it is typical to simplify things by correlating uniform voltage levels to uniform SoC percentages, and we do so here. The cell currents during self-balancing will not be very large, thus the expected time to achieve balance can be much longer than observed using other methods. But when considering the typical daily use of an EV, there are many long periods of time throughout the day when the vehicle is neither in use nor charging, such as when the driver is at work and/or sleeping. Therefore, a lengthy balancing process can often be tolerated in a typical use case.

This grid-to-parallel reconfiguration scheme also facilitates balancing during charging, without requiring shunt bypassing systems, or used in conjunction with such systems. Thus, both activities of balancing during rest periods and during charging are investigated in this article. In the next section, the experimental testbed employed is described, followed by a section describing the testing methodologies used. Then results for both cases of balancing and charging in the grid-to-parallel reconfiguration are presented, and discussion is provided.

II. EXPERIMENTAL TESTBED

Automotive grade ANR26650 $LiFePO_4$ cells from A123 Systems serve as the test cells for experiments in this study. These cells have a nominal capacity of 2.3 Ah, and a nominal open-circuit voltage (OCV) of 3.3 V.

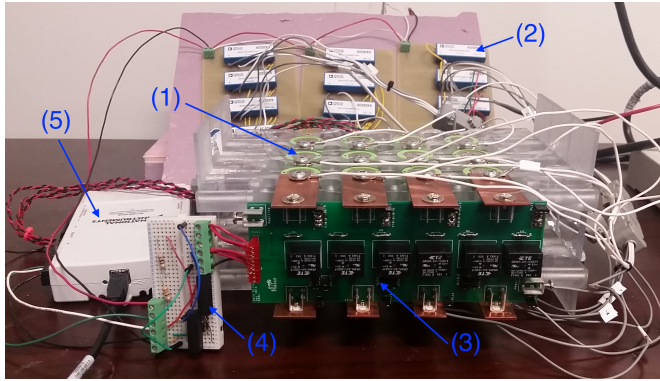


Fig. 2. Test Bed with (1) ANR26650 battery cells, (2) AD202KN Isolation Amplifiers, (3) Reconfiguration Circuit, (4) Relay Driving Circuit, (5) National Instruments 6218 DAQ.

The testbed for this investigation is a special-purpose research pack designed for table-top use, where multiple configurations are possible. Individual parallel modules of the grid configuration packs are connected by copper bars. A circuit-board mounted on the side of the pack controls the series/parallel connections between the modules via six automotive relays controlled by a National Instruments 6218 data acquisition unit (DAQ), in conjunction with LabVIEW. A photo of the setup is shown in Figure 2.

Voltage measurements were recorded by the 6218 DAQ and LabVIEW by means of Analog Devices AD202KN isolation amplifiers. Although current measurement in each

cell is possible through shunt resistors and amplification instrumentation incorporated into this testbed for various other studies [11][12][13], due to the expected low cell currents these current sensors were ineffective for this investigation. That is, shunts with enough resistance to result in a sufficient signal-to-noise ratio would detrimentally affect the process of self balancing. SoC was instead obtained from voltage measurements using

$$SoC(u) = 0.6867 \tan^{-1}(36.5475u - 120.3363) + 1.3017, \quad (1)$$

where u is the voltage; this equation results from a simple curve fit, with an R^2 value of 0.9827, from OCV-SoC reference point tests performed on the ANR26650 at 25°C [14]. The tests consisted of discharging the cell in 10% SoC increments followed by a 60 minute rest period between each pulse, to obtain the OCV at each SoC level.

Further comments on Equ. (1) and its use are in order. Inclusion of the SoC as a measure of energy imbalance is a convenient but not entirely necessary way to characterize results of this scheme. That is, one concern might be with the accuracy of using Equ. (1) because the fit is only valid for a given point in time (that is, the OCV and SoC are not updated or calculated in real time). However, it is not unreasonable to envision a scenario wherein Equ. (1) might be updated periodically, for example, during routine maintenance or pack characterization during downtime.

III. TEST METHODOLOGY

A. Balancing at Rest

Three pack configurations — 4S1P, 4S2P, and 4S3P — were investigated over different balancing durations and different initial SoCs to ascertain relationships between initial state and balancing performance over time. Using three grid configurations of increasing complexity in this manner not only allowed us to progress through verification of the experimental process, but also allowed us to investigate any possible differences in pack size (number of modules) on the configuration scheme as related to other observations. The packs were observed over the course of one, two, and three hour periods to determine the effect of time duration on improvement in SoC balance. These nine sequences were performed twice, first with initial cell SoCs between 50% and 10%, and again with SoCs between 100% and 50%.

Each test iteration began with the pack resting in grid-configuration for ten minutes to attain each cell's initial SoC. The grid-to-parallel reconfiguration is then activated via a relay network for the balancing time of that iteration. After balancing, the pack is switched back to grid configuration and rests for one hour to obtain the OCV, and using Equ. (1), the SoC of each cell.

In tabulated data to follow, pack imbalance is quantified by the range of cell energy in Ah, which is a scalar multiple of SoC. The typical metric looked at is the “imbalance” quantified in terms of

$$|SoC_{max} - SoC_{min}| \quad (2)$$

where SoC_{max} represents the cell with maximum SoC in the parallel configuration, and SoC_{min} represents the cell with the minimum SoC. The performance of each self-balancing test is quantified by the change in pack imbalance over the course of the test, normalized by the initial imbalance (in terms of percentage).

B. Balancing While Charging

The same equipment and setup used in the balancing tests of the previous subsection are used to compare charging configurations. The 4S3P pack was initialized with cell voltages given in Table I, charged at a constant voltage in three trials. During the first trial, the pack was charged in grid configuration at a constant voltage (CV) of 13.8 V until a cell reached the threshold voltage – 3.475 V for these tests. The second trial had the pack charging in parallel configuration at 3.45 V – 13.8 divided by the number of module in series – for the same amount of time as trial one, so as to isolate the effect of pack configuration on the charging process. During the third trial, the pack charged in parallel configuration at 3.5 V for 210 minutes to elucidate the benefits of charging in parallel configuration. For each trial, the pack rested for three hours to allow the terminal voltages to settle and provide an accurate OCV for SoC calculation.

TABLE I
INITIAL CELL VOLTAGES

	V1A-C	V2A-C	V3A-C	V4A-C
Grid	3.307 V	3.317 V	3.308 V	3.328 V
Parallel 1	3.307 V	3.328 V	3.307 V	3.317 V
Parallel 2	3.307 V	3.317 V	3.308 V	3.328 V

IV. RESULTS AND DISCUSSION

A. Balancing at Rest: High SoC Packs

The observed balancing performance of the packs tested with initial cell SoC's above 50% SoC is shown below in Table II and Figure 3. in this table (and tables to follow), the percentage imbalance is expressed using Equ. (2) in terms of Ah to emphasize energy imbalance. The majority of the improvement in pack balance occurs within the first hour, and tends to decrease logarithmically as balancing time increases. Although our goal was not focused on finding differences due to the number of modules in series for a grid configuration pack, we concluded from this limited number of experimental trials that there appears to be no clear correlation between pack size and balancing performance.

As an example of the calculations leading to the information in Tables II and III, suppose the initial terminal voltage for the k th cell in the j th module is represented as $v_i^{(j,k)}$, whereas the final terminal voltage is represented as $v_f^{(j,k)}$. Initial ($S_i^{(j,k)}$) and final ($S_f^{(j,k)}$) SoCs are calculated using Equ. (1) as

$$S_i^{(j,k)} = SoC(v_i^{(j,k)}) \quad , \quad S_f^{(j,k)} = SoC(v_f^{(j,k)}) \quad . \quad (3)$$

Using Equ. (2), initial and final pack imbalance – E_i and E_f , respectively – for the trial can be calculated in units of

Ah. Expressing these imbalances as percentages is a simple matter of dividing by the nominal capacity of the cells (2.3 Ah). The value for “Improvement” in the tables is then calculated as described previously, or

$$\frac{E_i - E_f}{E_i} \times 100 \quad (\%) \quad . \quad (4)$$

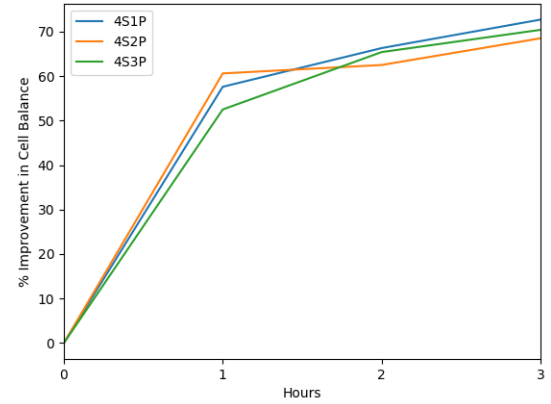


Fig. 3. Balancing Performance vs. Time: 100% - 50% SoC packs

TABLE II
IMBALANCE DATA: HIGH INITIAL SoC

1 Hour Balancing Time			
Config.	Init. Imbalance	Final Imbalance	Improvement
4S1P	0.613 Ah (26.6%)	0.260 Ah (11.2%)	57.6%
4S2P	0.772 Ah (33.6%)	0.304 Ah (13.2%)	60.6%
4S3P	0.604 Ah (26.3%)	0.288 Ah (12.5%)	52.5%
2 Hour Balancing Time			
Config.	Init. Imbalance	Final Imbalance	Improvement
4S1P	0.668 Ah (29.0%)	0.225 Ah (9.8%)	66.3%
4S2P	0.646 Ah (28.1%)	0.242 Ah (10.5%)	62.5%
4S3P	0.665 Ah (28.9%)	0.230 Ah (10.0%)	65.4%
3 Hour Balancing Time			
Config.	Init. Imbalance	Final Imbalance	Improvement
4S1P	0.384 Ah (16.7%)	0.105 Ah (4.6%)	72.7%
4S2P	0.617 Ah (26.8%)	0.194 Ah (8.4%)	68.5%
4S3P	0.672 Ah (29.2%)	0.198 Ah (8.6%)	70.4%

B. Balancing at Rest: Low SoC Packs

In the upper-half of the SoC range, the ANR26650 OCV curve is flatter than in the lower half, meaning a given change in cell voltage signifies a smaller change in SoC at a high SoC than at a low SoC. Therefore, it is expected that cells with low initial SoCs will balance more efficiently than those with high initial SoCs. We also note that in running these tests initial imbalances are typically higher for low SoC packs than for the case of high initial SoC packs. The reason for this is simply a consequence of the testing procedure. That is, when cells are intentionally discharged for these types of tests, often times at high C-rates to minimize time for testing, in the low SoC range the drop off in cell voltage is significantly faster and more challenging to gauge. Therefore, when random imbalances are established prior to

the test, a larger range of initial imbalances often results. This phenomenon is seen when comparing the tabulated data in Table II and Table III to follow.

The observed balancing performance of the packs tested with initial cell SoC's above 50% SoC is shown in Table III and Figure 4. The logarithmic trend in balancing performance over time mimics that of the high SoC packs, though as was expected, the performance is nearly 20% higher at low SoC. Each pack configuration improved cell balance by over 90% after three hours, compared to the 70% improvement seen in the high SoC packs.

Unlike the high SoC packs, the low SoC packs display a pattern in the self-balancing time characterization. In the first hour, balancing improvement increases with the number of cells in parallel. In the second and third hours, the pattern reverses. The 4S x P packs increase in effective capacitance by a factor of x . With x more sources, the initial throughput is higher than smaller packs. But the higher capacitance increases the settling time of the equivalent circuit network.

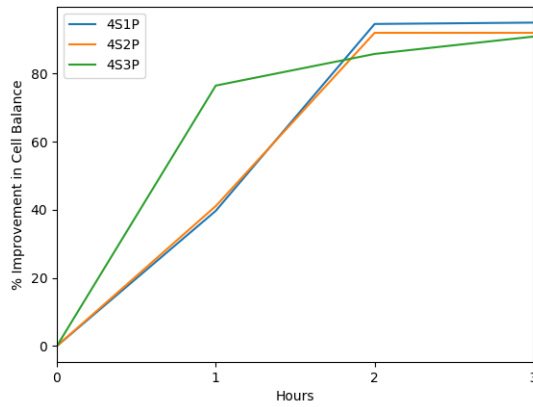


Fig. 4. Balancing Performance vs. Time: 50% - 10% SoC packs

TABLE III
IMBALANCE DATA: LOW INITIAL SoC

1 Hour Balancing Time			
Config.	Init. Imbalance	Final Imbalance	Improvement
4S1P	0.956 Ah (41.6%)	0.577 Ah (25.1%)	39.7%
4S2P	0.800 Ah (34.8%)	0.471 Ah (20.5%)	41.1%
4S3P	0.892 Ah (38.8%)	0.210 Ah (9.1%)	76.5%
2 Hour Balancing Time			
Config.	Init. Imbalance	Final Imbalance	Improvement
4S1P	0.893 Ah (38.8%)	0.048 Ah (2.1%)	94.6%
4S2P	0.803 Ah (34.9%)	0.063 Ah (2.7%)	92.0%
4S3P	0.777 Ah (33.8%)	0.110 Ah (4.8%)	85.8%
3 Hour Balancing Time			
Config.	Init. Imbalance	Final Imbalance	Improvement
4S1P	0.888 Ah (38.6%)	0.044 Ah (1.9%)	95.0%
4S2P	0.883 Ah (38.4%)	0.071 Ah (3.1%)	92.0%
4S3P	0.773 Ah (33.6%)	0.070 Ah (3.0%)	90.9%

C. Balancing at Rest: Use Case Simulation

To evaluate the overall utility of the self-balancing method, a use case simulation was performed on the 4S3P battery

TABLE IV
FINAL PACK CONDITIONS IN SIMULATION

	Voltage	SoC
Maximum	3.337 V	2.005 Ah
Minimum	3.334 V	1.981 Ah
Difference		0.024 Ah (1.04%)

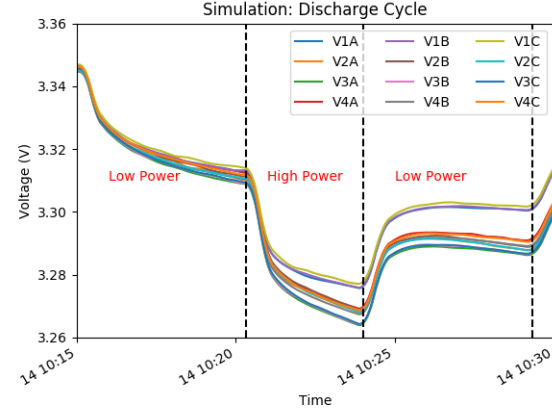


Fig. 5. Cell voltage during discharge cycle

pack. The use case selected for simulation was that of an EV with 2,880 cells – such as the Tesla Model S – used to commute 15 minutes to work and remain in the parking lot for the 7 to 9 hours the driver is engaged at work. For the commuting period, the battery pack was initially balanced – defined in this case as less than 2% imbalance in the pack – set to grid configuration, and discharged in three stages. The first and third stages consisted of discharging across a 6.5Ω load for 5 minutes each. This correlates to roughly the same cell-level current discharge as would be seen by a 2,880 cell pack delivering 5 kW to the motor, for maintaining speed. The second stage discharged the pack across a 2Ω load for 4 minutes, correlating to roughly 30 kW of power in the 2,880 cell pack for accelerating the EV. One minute of rest was included at the end of the third discharge stage. After the 15 minute commute ended, the pack was reconfigured to parallel, and left to self-balance for seven hours, after which the pack was reconfigured to grid configuration and allowed to rest for two hours to determine the final OCV and SoC imbalance of the pack. The results of the simulation are shown in Table IV and Figure 5 - 6. During the discharge periods of the simulation, illustrated in Figure 5, the voltage drops of the cells are not uniform. This indicates the pack is becoming less balanced as the drive continues. However, when given 7 hours to self-balance, the battery pack shows a 0.024 Ah imbalance, which when normalized by the capacity yields 1.04%, satisfying the conditions of pack balance.

D. Balancing While Charging

Returning now to the experimental testbed of this study, the initial imbalance, final imbalance, and percent change in imbalance for each trial are given in Table V. When charged

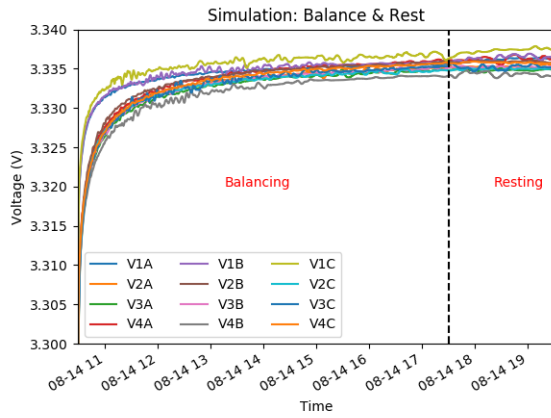


Fig. 6. Cell voltage during pack balancing and resting

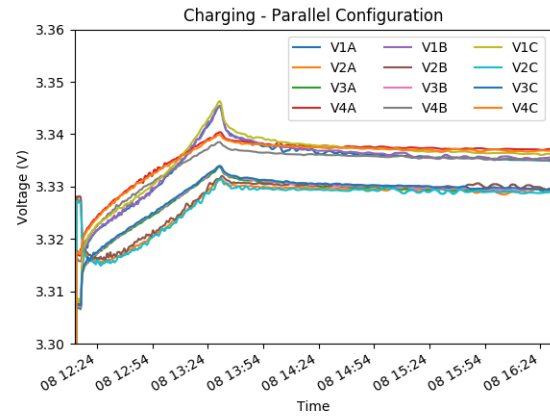


Fig. 8. Cell voltage during charging – 4S3P Pack in parallel configuration

in grid configuration at CV of 13.8 V, the terminal voltages of the pack cells were recorded as depicted in Figure 7. The initial imbalance in the pack decreased by 73.3% with the pack in grid configuration. When charged for the same amount of time at a CV of 3.45 V in parallel configuration, the pack balance improves by 78.7% — slightly better than the grid configuration trial. In this trial, the pack did not charge as much as the grid configuration pack. But it can be seen in Figure 8 that the cell voltages increase more evenly. This suggests that when the pack is in parallel configuration, it can charge for a longer time without cells reaching a threshold voltage. This idea was tested in the third trial, and results are shown in Figure 9. When the pack is charged in

in grid configuration until a cell reaches a threshold voltage V_{TH} , followed by switching to CV at V_{TH} with the pack switched to parallel configuration. This will prevent cells from overcharging while lower SoC cells can continue to charge, without wasting energy through shunt resistors.

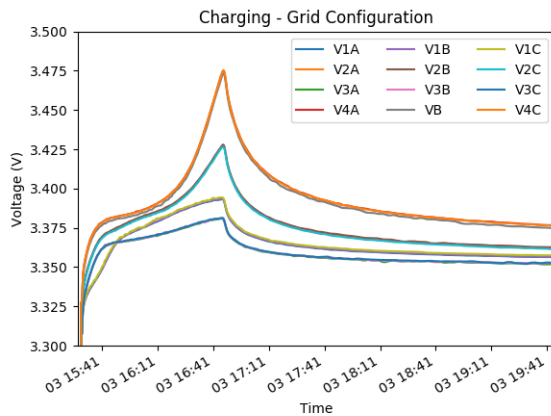


Fig. 7. Cell voltage during charging – 4S3P Pack in grid configuration

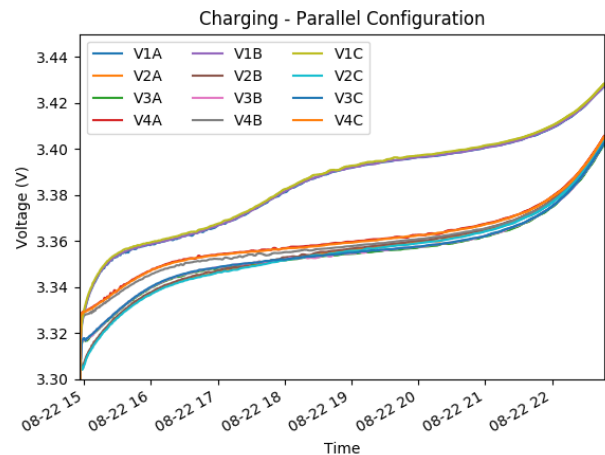


Fig. 9. Cell voltage during charging – 4S3P Pack in parallel configuration

TABLE V
CHANGE IN IMBALANCE DURING CHARGE

Config.	Init. Imbalance	Final Imbalance	Change
Grid	0.296 Ah (12.9%)	0.079 Ah (3.4%)	73.3%
Parallel 1	0.301 Ah (13.1%)	0.065 Ah (2.8%)	78.7%
Parallel 2	0.296 Ah (12.9%)	0.030 Ah (1.3%)	89.9%

V. CONCLUSIONS

This article describes an innovative idea for pack reconfiguration to improve cell balancing for grid configuration packs. The scheme has the simplicity of passive balancing with the energy efficiency characteristic of active balancing approaches. Through experimentation this approach has

been investigated for balancing at rest and balancing during charge. While it is true that results using the reconfiguration scheme presented here for balancing while at rest inherently require much more time than other schemes, this scheme does not suffer from drawbacks inherent with other passive and active balancing approaches, such as wasted energy (passive dissipation) or added complexity and additional electronics (active schemes). Moreover, most consumer uses of EVs are characterized by significant periods of downtime, which speaks favorably to the feasibility of the approach presented here.

For practical implementation, power MOSFETs should be utilized for the reconfiguration switches, because relays require continuous current to maintain the on-state, thereby increasing the parasitic power needed for the reconfiguration control. And while self-balancing currents will not be large, the switches will still need to conduct currents of magnitudes normally required in EVs. The reconfiguration method given by [15] can be used in this application as well, with minor adjustments to account for the parallel connectivity. Since the MOSFETs will switch at a frequency on the order of 50 μ Hz, switching losses will be negligible, thus thermal heating of the MOSFETs will be contained to only the effect of continuous current. Any thermal management required can be accomplished through the existing pack cooling system.

Reconfiguration of a battery pack from grid to parallel configuration can effectively improve the cell balance of the pack at rest more efficiently, and more simply, than other methods. This scheme does not require the complex control circuitry needed by switching converters, nor does it have the switching losses. And unlike passive balancing, which is competitively simple, self-balancing maintains the majority of net energy in the pack. The only source of energy dissipation is cell internal resistance and the resistance of the cell connections, factors that are inherent in any battery pack. The self-balancing characterization tests showed no clear relationship between pack size with balancing time and balancing performance, which points to a possible extension of these conclusions to larger packs with many modules. We also conclude it is possible that the performance of self balancing is sensitive to the initial range and distribution of cell SoC, factors for which this test setup was unable to adequately investigate. However, it was demonstrated that cell balancing was effective over time. The use case simulation demonstrated the effectiveness of self-balancing in situations where long periods of time pass between uses of the pack.

The use of grid-to-parallel pack reconfiguration for charging purposes also has clear benefits. The pack charged in parallel configuration achieved nearly 17% more balance than the pack charged in grid configuration. The grid configuration pack would need shunt resistors to bypass fully charged cells and continue to charge, prevent damage, and decrease imbalance, thereby drawing more energy than is delivered to the pack. Reconfiguring the pack to parallel for charging will take longer to fully charge the pack (though typical EV use inherently contains these periods), but will charge the

pack more uniformly, lead to a more balanced pack, and do so without excess power loss. To rigorously justify the use of this scheme in charging, the benefits achieved would have to be weighed against the additional cost of implementation (additional switches in the battery pack for reconfiguration).

Because it is evident that reconfiguration of battery packs can be an expensive proposition, future work in this area will involve additional experimentation to explore comparable timing and efficiency measurements for other configurations used in passively balanced packs. Current work is moving in this direction as well as to simulate (calculate) what a comparable passive balancing scheme would waste in energy on the testbed used here in order to further explore benefits of the proposed reconfiguration scheme.

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